

Directions: Write answers on this page and show any work on separate pages.

Problem 1: Compute the following indefinite integrals. In each case, assume n is constant. Show work by applying integration techniques. (Recall: $i = \sqrt{-1} \Rightarrow i$ is a constant)

$$(a) \int \sin(nx) dx = -\frac{1}{n} \cos(nx) + C$$

$$(b) \int e^{inx} dx = \frac{1}{in} e^{inx} + C$$

$$(c) \int x e^{inx} dx = \frac{x}{in} e^{inx} + \frac{1}{n^2} e^{inx} + C$$

$$(d) \int x^2 e^{inx} dx = \frac{x^2}{in} e^{inx} + \frac{2x}{n^2} e^{inx} - \frac{2}{in^3} e^{inx} + C$$

Problem 2: Determine all possible values of the expressions below for $n = 1, 2, 3, \dots$ by rewriting each expression as a piece-wise function of n or preferably as an equivalent single function of n . (Hint: Try some values of n to see a pattern)

$$(a) \sin(\pi n) = 0 \quad \cos(\pi n) = (-1)^n \quad e^{i\pi n} = (-1)^n$$

$$(b) \sin(2\pi n) = 0 \quad \cos(2\pi n) = 1 \quad e^{i2\pi n} = 1$$

$$(c) \sin\left(\frac{\pi}{2}n\right) = \begin{cases} 0 & n \text{ even} \\ -1 & n = 3, 7, 11, \dots \\ 1 & n = 1, 5, 9, \dots \end{cases} \quad \cos\left(\frac{\pi}{2}n\right) = \begin{cases} 0 & n \text{ odd} \\ -1 & n = 2, 6, 10, \dots \\ 1 & n = 4, 8, 12, \dots \end{cases} \quad e^{i\pi n/2} = i^4$$

Problem 3: Circle the correct answer in brackets [] and/or fill in any blanks.

$$(a) \text{ If } f(x) \text{ is an even function then } \underline{f(-x) = f(x)} \text{ (Look up the definition somewhere)}$$

$$(b) \text{ If } f(x) \text{ is an odd function then } \underline{f(-x) = -f(x)}$$

$$(c) \text{ If } f(x) \text{ is an even function then } \int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$$

$$(d) \text{ If } f(x) \text{ is an odd function then } \int_{-L}^L f(x) dx = 0$$

$$(e) \text{ Cosine is an [EVEN / ODD] function, therefore } \cos(-x) = \cos x$$

$$(f) \text{ Sine is an [EVEN / ODD] function, therefore } \sin(-x) = -\sin x$$

$$(g) \text{ By Euler's formula and (c) \& (d), } e^{-inx} + e^{inx} = 2 \cos(nx)$$

$$(h) \text{ By Euler's formula and (c) \& (d), } (e^{-inx} - e^{inx})i = 2 \sin(nx)$$